



COMPSCI 130 S1 2022

Introduction to Software Fundamentals

Linked Lists – Introduction



Agenda & Reading

- ▶ **Agenda**
 - ▶ Introduction

- ▶ **Online Coursebook:**
 - ▶ <https://onlinetextbook.auckland.ac.nz/l7-linked-lists/>



Review

- ▶ We have used Python lists to implement the abstract data types presented (Stack and Queue).
 - ▶ The list is a powerful, yet simple, collection mechanism that provides the programmer with a wide variety of operations.
- ▶ A Python list
 - ▶ stores each element in **contiguous memory** if possible. This makes it possible to access any element in $O(1)$ time.
 - ▶ However, insertion or deletion elements at the beginning of the list takes $O(n)$.



The List ADT

- ▶ A list is a collection of items where each item holds a **relative position** with respect to the others. We can consider the list as having a first item, a second item, a third item, and so on. We can also refer to the **beginning** of the list (the first item) and the **end** of the list (the last item).
- ▶ Unordered Vs Ordered
 - ▶ Unordered meaning that the items are not stored in a sorted fashion.

54, 26, 93, 17, 77 and 31

17, 26, 31, 54, 77 and 93

- ▶ A Python list ([]) is an implementation of an unordered list,



The List ADT

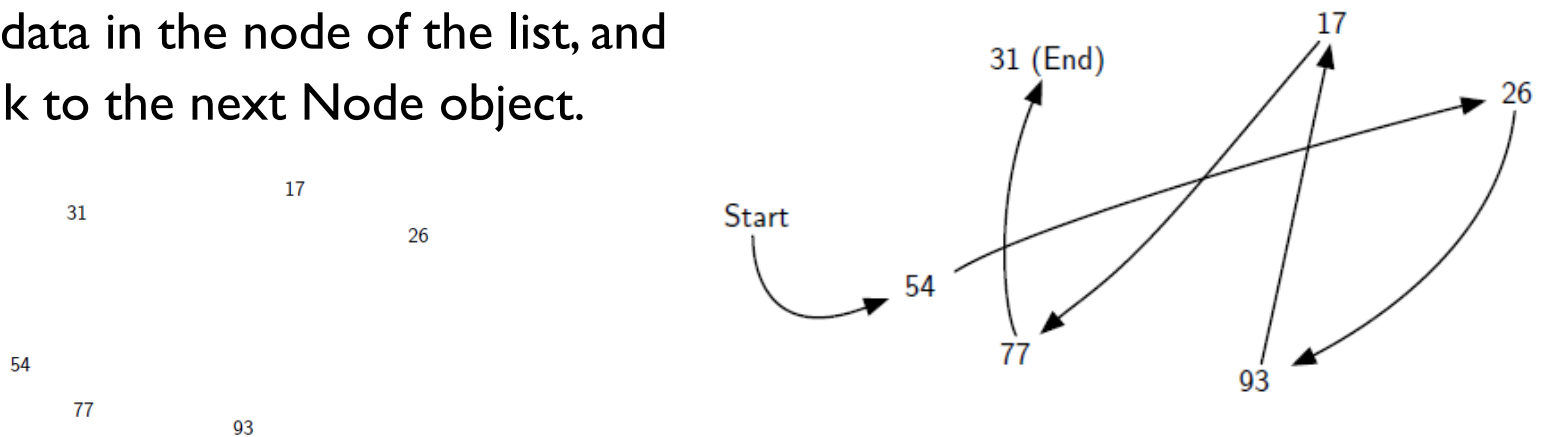
- ▶ What are the operations which can be used with a List Abstract Data?
- ▶ creates a new list that is empty.
 - ▶ It needs no parameters and returns an empty list.
- ▶ `add(item)` adds a new item to the list.
 - ▶ It needs the item and returns nothing.
- ▶ `remove(item)` removes the item from the list.
 - ▶ It needs the item and modifies the list. **Assume** the item is **present** in the list.
- ▶ `search(item)` searches for the item in the list.
 - ▶ It needs the item and returns a boolean value.
- ▶ `is_empty()` tests to see whether the list is empty.
 - ▶ It needs no parameters and **returns** a boolean value.
- ▶ `size()` returns the number of items in the list.
 - ▶ It needs no parameters and **returns** an integer.

No checking is
done in the
implementation!



Contiguous Memory

- ▶ A Python list stores each element in contiguous memory if possible.
- ▶ List ADT – there is no requirement that the items be stored in contiguous memory
- ▶ In order to implement an unordered list, we will construct what is commonly known as a **linked list**.
 - ▶ A Node object will store
 - ▶ the data in the node of the list, and
 - ▶ a link to the next Node object.





Insertion and Deletion

- Items can be inserted into and deleted from the linked list without shifting data.

